

GOTHAMITE is an **NIST framework driven** platform that ingests and correlates multi-source dark web and open-source traces to build an evidence-backed **actor intelligence graph**, providing actionable insights for investigators.

Proposed Solution



Multi-Source Intelligence Ingestion

Collects data from dark web, forums, marketplaces, paste sites and blockchain.



Identity Resolution & Linking

Maps aliases, accounts, wallets and infrastructure to same actor using **NIST framework**.



Actor Intelligence Graph

Combines textual, behavioural, temporal and graph signals.

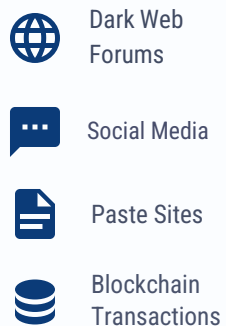


Investigator-in-the-Loop

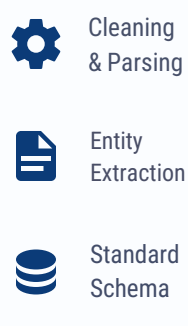
Provides explainable insights, source references and exportable reports.

Solution Workflow (NIST Framework)

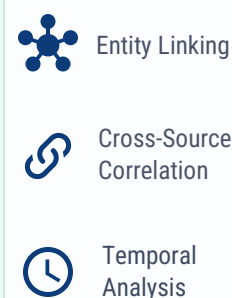
1 Data Collection (N)



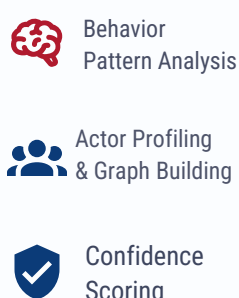
2 Integration (I)



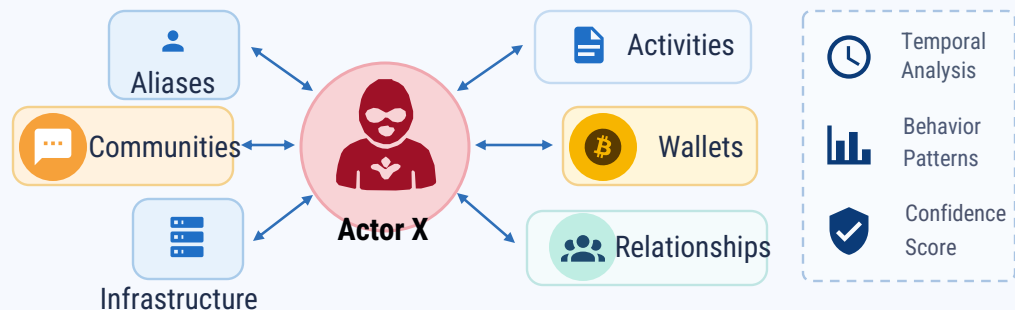
3 Standardization (S)



4 Threat Inference (T)



Actor Intelligence Graph (Output)



How It Addresses the Problem



Links fragmented identities across platforms.



Reduces manual investigation time with automated correlation.



Unified view of actor activities, infrastructure and financial traces.



Evidence-based insights with source references and confidence scores.

Innovation & Uniqueness (NIST Framework)

A

On Prem Support

B

Multi-Source Correlation

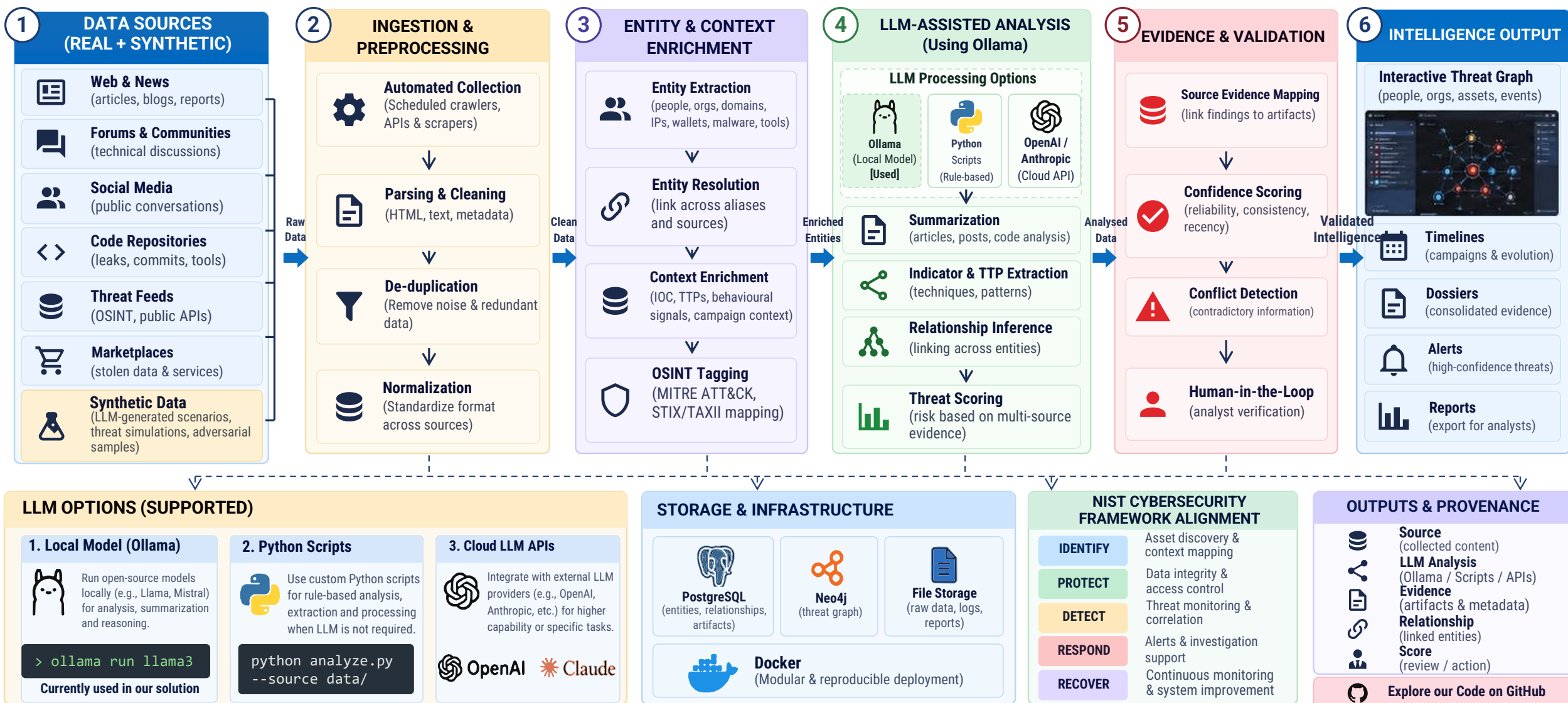
C

Dynamic & Explainable Graph

D

Investigator-Centric Design

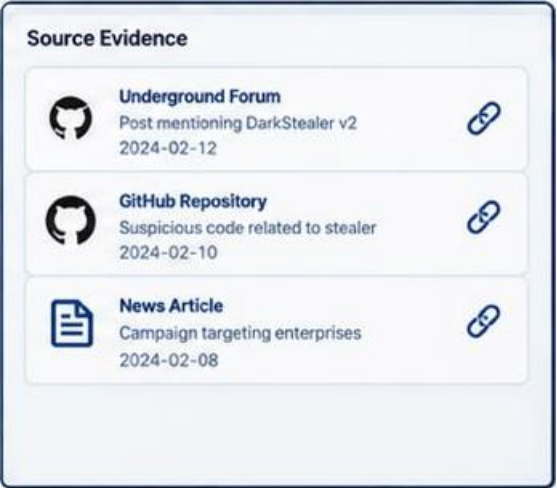
A modular pipeline that combines **real and synthetic data**, **OSINT**, and **LLM-powered analysis** to generate an explainable threat intelligence graph with evidence and confidence for proactive cyber defense.





Our solution is technically feasible, scalable, and cost-effective, using open-source tools and LLM-powered analysis to generate an explainable threat intelligence graph for real-world use.

PROTOTYPE SNAPSHOTS



FEASIBILITY ANALYSIS

Technical Feasibility Can be implemented using open-source tools and APIs.	- Built using Python, FastAPI, Neo4j, Ollama and other open-source tools. - Supports modular and scalable architecture. - Integrates multiple data sources (OSINT, social media, code repos, etc.).
Operational Feasibility Usable for real-world investigative workflows.	- Web-based dashboard for analysts. - Minimal training required. - Automated data processing reduces manual effort.
Economic Feasibility Cost-effective and sustainable.	- No costly proprietary tools required. - Deployable on standard cloud or on-premise servers. - Scales with incremental infrastructure as per need.
Legal & Ethical Feasibility Complies with legal frameworks and responsible data handling.	- Uses only publicly available and authorized sources. - Follows data privacy and cybersecurity guidelines. - Implements access control and audit logging.

POTENTIAL CHALLENGES AND RISKS

	Heterogeneous & Noisy Data Unstructured and frequently changing sources.
	Evasion by Threat Actors Use of VPNs, proxies, obfuscation techniques.
	Inaccurate or Weak Links False correlations due to sparse signals.
	Scalability & Performance Large volumes of multi-source data and graph computations.

STRATEGIES TO OVERCOME

	Robust Data Processing Custom parsing pipelines, deduplication, noise filtering and schema normalization.
	Multi-Signal Correlation Combine textual, behavioural, temporal and graph signals.
	Confidence Scoring & Validation Use multi-source evidence with human-in-the-loop verification.
	Modular & Scalable Architecture Distributed processing, efficient graph database (Neo4j) and incremental data pipelines.

VALUE PROPOSITION & IMPACT

Faster & More Accurate Threat Detection	Reduced Manual Investigation Effort	Scalable and Cost-Effective Solution	Supports Proactive Cyber Defense	Adaptable to Multiple Domains and Use Cases
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1 KEY STAKEHOLDERS (Target Audience)



Law Enforcement & Investigative Agencies



National Security & Intelligence



**Cybersecurity Organizations
(CERT-In, NCIIPC)**



Research Community



Policy Makers & Regulators



General Public

2 IMPACT ON TARGET AUDIENCE



SOCIAL IMPACT

Safer digital environment,
reduced cyber frauds,
increased awareness and trust.



ECONOMIC IMPACT

Lower investigation cost,
efficient resource allocation,
minimizes financial losses.



OPERATIONAL IMPACT

Faster, evidence-backed
investigations with unified
actor intelligence.



ENVIRONMENTAL IMPACT

Leverages existing digital
infrastructure and open-source
tools, reducing redundant systems.

3 COMPARISON WITH EXISTING SOLUTIONS

Aspect	Existing Solutions	GOTHAMITE (Proposed)
Data Coverage	Single-source (e.g., only dark web or only OSINT)	Multi-source (dark web, OSINT, blockchain, forums, breach DBs)
Correlation	Keyword / rule- based limited linkage	NIST-guided multi-layer correlation (text, behaviour, temporal, graph)
Actor Resolution	Often partial or inaccurate	Alias and persona linking with confidence scoring
Evidence Handling	Limited provenance	Source references, audit trail and exportable reports
Standards Compliance	Rarely aligned with formal frameworks	Aligned with NIST CSF, SP 800-53/61/86, MITRE ATT&CK, STIX/TAXII
Deployment	Closed / high cost	Open-source, modular, scalable (cloud/on-premise)

4 FUTURE SCOPE & LONG TERM VALUE

1

Integrate additional sources
(social media, IoT, cloud logs).

2

Real-time threat monitoring
and alerting.

3

Advanced graph analytics
and predictive actor modelling.

4

Collaboration with national
and international threat intel
platforms.

5

Continuous improvement
with feedback loop from
investigators.

OVERALL IMPACT



Stronger National Security

Proactive cybercrime and
terrorism mitigation



Informed Decision Making

Evidence-backed,
faster investigations



Greater Public Trust

A safer and more resilient
digital society



Sustainable and Scalable

Open-source, standards-based
solution for long-term use

RESEARCH AND REFERENCES

Our solution is based on established cybersecurity frameworks, threat intelligence research and open-source resources to ensure a reliable and standardized approach.

1 STANDARDS & FRAMEWORKS

NIST Cybersecurity Framework (CSF 2.0)	https://www.nist.gov/cyberframework
NIST SP 800-53	https://csrc.nist.gov/publications/detail/sp/800-53/final
NIST SP 800-61	https://csrc.nist.gov/publications/detail/sp/800-61/rev-3/final
NIST SP 800-86	https://csrc.nist.gov/publications/detail/sp/800-86/final
MITRE ATT&CK	https://attack.mitre.org/
STIX 2.1 / TAXII 2.1	https://oasis-open.github.io/cti-documentation/
OWASP	https://owasp.org/

2 RESEARCH PAPERS

Darknet and Deepnet Mining for Proactive Cybersecurity	https://arxiv.org/abs/1607.08583
Graph-based Methods for Cyber Threat Intelligence	https://arxiv.org/abs/2101.00459
Stylometry for Author Identification (Text Analysis)	https://arxiv.org/abs/1703.09256
Entity Resolution in Cybersecurity	https://arxiv.org/abs/2004.07888
OSINT for Threat Intelligence	https://arxiv.org/abs/2005.03682
Behavioral Analysis of Online Actors	https://arxiv.org/abs/1904.08040

3 DATA / TECHNICAL SOURCES

Dark Web / OSINT Sources	https://osintframework.com
Threat Intelligence Feeds	https://otx.alienvault.com
Breach and Incident Datasets	https://haveibeenpwned.com
Malware & Indicator Repositories	https://www.virustotal.com
Tor and Hidden Services Research	https://metrics.torproject.org/
Blockchain / Wallet Analysis	https://chainanalysis.com/blog/
Cyber Threat Reports	https://www.mandiant.com/resources

4 PROJECT / IMPLEMENTATION REFERENCES

GOTHAMITE GitHub Repository	https://github.com/Shankh-AvivCREW/GOTHAMITE
Project Documentation	https://github.com/Shankh-AvivCREW/GOTHAMITE/wiki

REFERENCE FLOW

Standards & Frameworks



Research Papers



Data Sources



Project Repository & Documentation